Unit 4 Progress Check: FRQ

Name

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

$$\mathrm{HCl}(aq) + \mathrm{NaOH}(aq) \rightarrow \mathrm{NaCl}(aq) + \mathrm{H_2O}(l)$$

A student was given the task of titrating a $20.\ \mathrm{mL}$ sample of $0.10\ M\ \mathrm{HCl}(aq)$ with $0.10\ M\ \mathrm{NaOH}(aq)$. The $\mathrm{HCl}(aq)$ was placed in an Erlenmeyer flask. An equation for the reaction that occurs during the titration is given above.

(a) According to the equation for the reaction, if the amount of the reactants is halved, how does this affect the amount of $\mathrm{H_2O}(l)$ produced in the reaction?

Please respond on separate paper, following directions from your teacher.

(b) The equation above is not written in net ionic form. Write the correct net ionic equation for the reaction.

Please respond on separate paper, following directions from your teacher.

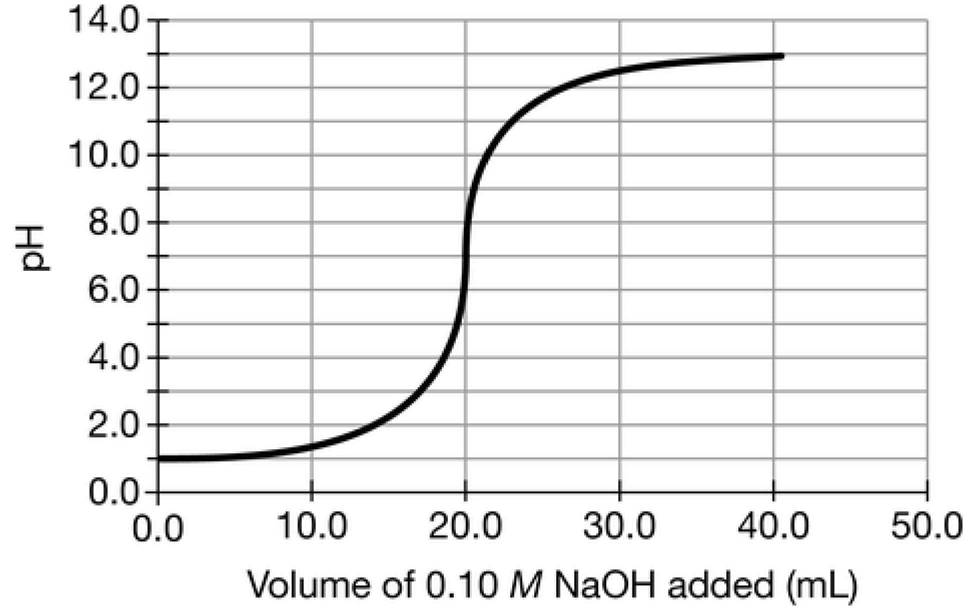
(c) In the equation you wrote in part (b), which species is a proton acceptor?

Please respond on separate paper, following directions from your teacher.

(d) How many moles of HCl are in the $20.\ \mathrm{mL}$ sample of $0.10\ M\ \mathrm{HCl}(aq)$?

Please respond on separate paper, following directions from your teacher.

Unit 4 Progress Check: FRQ

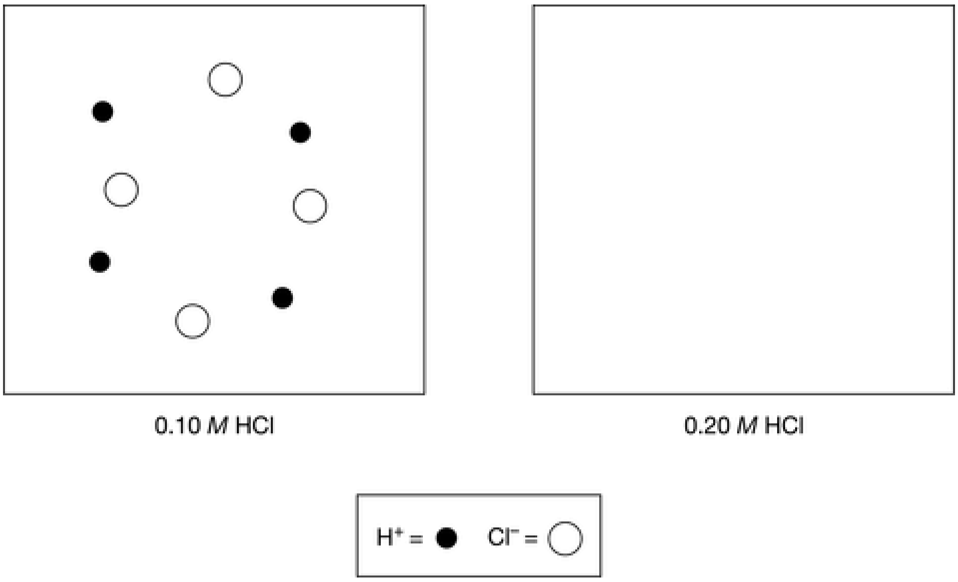


(e) The graph above shows the results of the titration of the $20.\ \mathrm{mL}$ sample of $0.10\ M\ \mathrm{HCl}(aq)$ with $0.10\ M\ \mathrm{NaOH}(aq)$. On the graph, draw an $\mathbf{X}$ to identify the location of the equivalence point of the titration.

Please respond on separate paper, following directions from your teacher.

The student performs a second titration using the $0.10\ M\ \mathrm{NaOH}(aq)$ solution again as the titrant, but this time with a $20.\ \mathrm{mL}$ sample of $0.20\ M\ \mathrm{HCl}(aq)$ instead of $0.10\ M\ \mathrm{HCl}(aq)$.

(f) The box below to the left represents ions in a certain volume of $0.10\ M\ \mathrm{HCl}(aq)$. In the box below to the right, draw a representation of ions in the same volume of $0.20\ M\ \mathrm{HCl}(aq)$. (Do not include any water molecules in your drawing.)



Please respond on separate paper, following directions from your teacher.

(g) On the graph in part (d), carefully draw a curve that shows the results of the second titration, in which the student titrates a $20.$ mL sample of $0.20\ M\ HCl(aq)$ with $0.10\ M\ NaOH(aq)$.

Please respond on separate paper, following directions from your teacher.

(h) The student made observations related to the contents of the Erlenmeyer flask during the titration. Identify an observation that could have led the student to conclude that a chemical change took place during the titration.

Please respond on separate paper, following directions from your teacher.